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Full Length Article

Measuring network-level internet censorship: DNS and IP-based filtering across Iraqi residential ISPs

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A B S T R A C T

Internet filtering is increasingly implemented through telecommunications operators, yet fragmented national network architectures complicate both enforcement and accountability. This study examines Iraq, where most ISPs fall under the Ministry of Communications, while the Kurdistan region operates under separate regulations, resulting in a fragmented network infrastructure. Active measurements from four Iraqi residential networks (three fixed ISPs and one mobile operator) were conducted over two month-long campaigns in 2025. These measurements tested 50,000 popular domains from the (Majestic Million) list and more than 100,000 domains from an official Ministry test list. DNS interference was detected using ZDNS; IP-level reachability was assessed with Nmap; domain-to-IP mapping via IPinfo hosted-domains data; and content classification was carried out with FortiGuard. DNS interference emerged as the dominant and most consistently deployed mechanism: across the popular-domain list, 386 domains were blocked on all four networks, including highly ranked domains. IP-level blocking proved less uniform: 195 blocked IP addresses appeared common across networks, though one large ISP enforced many additional blocks on IPs and domains. This led to substantial over-blocking, with collateral impacts concentrated in business, information-technology, and education categories due to shared hosting. The results demonstrate that centralized policy directives often yield uneven ISP implementation, highlighting concerns over transparency, proportionality, and the economic side-effects of broad blocking rules.

1. Introduction

Internet censorship refers to the intentional impairment or blocking of access to some online resources and services, usually enforced by governments or organizations, regardless of the intent, scope, or legitimacy of the actions (Hall et al., 2023). Internet censorship is an increasingly important topic of public debate worldwide. The way censorship is interpreted and enforced affects fundamental concepts like freedom of expression and national security. Variations in perspectives on censorship among individuals, societies, cultures, and legal systems reveal the core values of the people concerned (Master, 2023). From a technical perspective, Internet censorship can impact the overall Quality of Service (QoS), manifested as increased packet losses, and increased delays (Aceto and Pescapé, 2015). The enforcement of Internet censorship can be due to political motivations such as inhibiting specific groups or parties within a particular jurisdiction (Chaabane et al., 2014), or due to religious motivations (Pearce et al., 2018), while ethical considerations (Fletcher and Hayes-Birchler, 2023) focus on safeguarding children from unsuitable content. Internet censorship has resulted in significant economic impacts, as reported by the Internet Society. An estimated loss of $75,002,906 in combined Gross Domestic Product (GDP) has occurred across all nations that have undergone Internet shutdowns in the last year (2025) (Internet Society Pulse 2026). The issue of Internet censorship has gained prominence in recent years, with numerous countries implementing censorship and regulations to achieve particular objectives. As a result, the utilization of circumvention tools is on the rise; for example, the average number of users on the global Tor network has experienced a significant increase over the past five years. Between 2016 and 2021, the Tor network saw an average of 2 million users worldwide, with a peak reaching 4.5 million users. While an average of 3.2 million users have used the Tor network globally from 2021 to 2026, reaching a peak of 19.3 million users (Tor Project 2026).

National-scale Internet censorship can be categorized as either centralized or decentralized. In the case of centralized censorship, the

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degree of censorship is uniform across all Internet Service Providers (ISPs) within a given jurisdiction (Master, 2023, Xue et al., 2024, Sundara Raman et al., 2020), covering both the methods employed and the categories of content that are censored. In contrast, decentralized censorship is characterized by varying levels of censorship among ISPs operating within the same jurisdiction. This disparity arises from the differing network topologies and control points. Certain national network designs naturally serve as more effective choke points and control centers (Hall et al., 2023). Notable examples of centralized censorship include China (Sundara Raman et al., 2020) and Iran (Singh et al., 2020). On the other hand, countries like Russia (Sundara Raman et al., 2020), India (Master, 2023), and Iraq (Al-Musawi et al., 2025) exhibit decentralized censorship. Although centralized censorship has been extensively studied in prior work, decentralized censorship remains less explored. This gap can be attributed to the heterogeneity of ISP ecosystems that underpin decentralized systems. Such diversity necessitates multiple vantage points to obtain a comprehensive and accurate view of the censorship landscape (Ramesh et al., 2020). In light of this framework, the present paper evaluates the extent of Internet censorship in fragmented, decentralized networks, with Iraq serving as a focal case study.

The research by (Al-Musawi et al., 2025) reveals that Iraqi ISPs are under the governance of the Ministry of Communications (MoC). Consequently, their traffic is directed through a pivotal point for governmental oversight and censorship, which is managed by the ministry itself. Nevertheless, the findings suggest that this regulatory framework does not extend to ISPs operating in the Kurdistan Region, located in northern Iraq (The Kurdistan Regional Parliament 2026). The region functions autonomously under its own regulatory framework, resulting in a fragmented telecommunications and network infrastructure across Iraq. In the Kurdistan region, the Ministry of Transport and Communications (MoTaC) oversees the provision of landline and Internet services through joint-share arrangements with private telecommunications companies. Consequently, all ISPs operating within the Kurdistan region are required by MoTaC to comply with its regulatory directives (Kurdistan24 2026).

In this paper, we use the term decentralized to describe Iraq's network for two principal reasons. First, and most importantly, a substantial number of ISPs in the federal region do not, in practice, route their traffic through the MoC-mandated choke point. Instead, they route traffic through Kurdistan-based ISPs or establish direct international peering connections, often to avoid tax obligations (Al-Musawi et al., 2025). Second, the Kurdistan region operates under an autonomous policy framework and is not required to route traffic through the federal choke point. Collectively, these factors result in a fragmented network in which censorship policies are enforced inconsistently across ISPs (Khayyat and Muhamad, 2023).

Previous studies have utilized censorship data collected from censorship measurement platforms, such as Open Observatory of Network Interference (OONI) (Ververis et al., 2021) and Censored Planet (Raman et al., 2023). Nonetheless, these platforms might not reflect the true state of censorship due to the trade-offs between details and extent of coverage, as stated by (Raman et al., 2023, Master and Garman, 2023, Hoang et al., 2024). Other studies have utilized the Border Gateway Protocol (BGP) data to pinpoint the Autonomous Systems (ASes) that are responsible for instigating censorship activities (Al-Musawi et al., 2017, Levett et al., 2024).

To mitigate risks to human participants, researchers often rely on remote vantage points, such as SOCKS proxies (Jin et al., 2021), Virtual Private Servers (VPSes) (Wu et al., 2025), and Virtual Private Networks (VPNs) (Niaki et al., 2020). However, these vantage points may experience different or inconsistent censorship behavior, as they are typically hosted in commercial data centers (Niaki et al., 2020). In contrast, local vantage points, such as residential networks, more accurately reflect the censorship conditions experienced by end users. Due to limited direct access to such networks, researchers frequently rely on residential proxies or VPN-based access methods, which may still introduce bias. This paper examines the extent of Internet censorship in Iraq's federal region, excluding Kurdistan.

This study presents the first assessment of Internet censorship in Iraq, focusing on content filtering. Previous work has documented only sporadic shutdowns, without examining filtering methods. To overcome vantage point limitations, active measurements were conducted from residential networks to capture the censorship encountered by local citizens. The measurement framework included two test lists: a public test list derived from the Majestic Million domains (Majestic 2025), and a private test list from the Iraqi MoC. The study commences with domain-level measurements of Domain Name System (DNS) resolution, followed by network-level analysis in which domains are mapped to IP addresses using IPinfo's database. The process yielded a detailed set of target IPs, from which insights were derived regarding affected domains, censored categories, and ranking patterns.

This paper makes five primary contributions:

1. This study provides the first assessment of the level of Internet censorship in Iraq.
2. The limitations of remote vantage points are addressed through residential network measurements in Iraq.
3. Alongside the public domain test list, an exclusive domain list obtained from the Iraqi MoC is incorporated to represent government-imposed domains.
4. Our analysis reveals the impact of TCP/IP blocking on useful domain names within the Business and Information Technology sectors.
5. Despite the existence of the MoC regulatory framework, notable inconsistencies across ISPs indicate that its enforcement remains incomplete within Iraq's federal region.

This paper is structured in the following manner: Section 2 offers background information regarding Internet censorship, while Section 3 gives an overview of related research. Section 4 delves into our methodology for performing measurements, followed by Section 5, which elucidates our system design. Section 6 reports the results acquired, and ultimately, Section 7 discusses key implications from the obtained results. Finally, Section 8 presents the conclusion along with future directions.

2. Censorship background

Internet censorship refers to the control, suppression, or regulation of access to information and communication on the Internet by governmental, corporate, or institutional entities. It encompasses a range of practices aimed at restricting or manipulating the availability, visibility, or dissemination of digital content, often justified by concerns related to national security, public morality, political stability, intellectual property, or cultural norms (Amich et al., 2022).

Censorship mechanisms can be utilized to analyze various directional traffic, specifically, inbound and outbound censorship. Inbound censorship pertains to the monitoring and interception of traffic that flows into a network from external sources, such as those outside the country or region. Conversely, outbound censorship involves the examination of traffic that originates from within the network and is directed towards external destinations (Jin et al., 2025). Consequently, censorship can be categorized as either unidirectional or bidirectional (Bock et al., 2020). Unidirectional censorship refers to a filtering system that functions in a single direction of network communication, whether that be inbound or outbound. In contrast, bidirectional censorship encompasses filtering systems that operate in both directions concurrently. An example of unidirectional censorship is Iran's protocol filter system

(Bock et al., 2020). Other censorship frameworks, like those in Russia (Ramesh et al., 2023), specifically filter inbound traffic from certain countries aimed at governmental websites while also imposing restrictions on outbound traffic (Xue et al., 2022). This combined strategy illustrates bidirectional censorship.

Internet censorship can be enforced using various technical methods including Transmission Control Protocol (TCP) / Internet Protocol (IP) blocking, Hypertext Transfer Protocol (HTTP) / Uniform Resource Locator (URL) filtering, and DNS-based censorship. Table 1 summarizes our comparative analysis of the censorship methods based on five key aspects. The first aspect shows how traffic is identified for each censorship method, while the second reveals the possible interference techniques. The third and fourth aspects are the censor-oriented advantages and disadvantages. The fifth aspect describes the client-observable effects (also referred to as symptoms). TCP/IP blocking is relatively inexpensive and straightforward to implement; however, it may lead to significant over-blocking since numerous services can be hosted on a single web server (Hall et al., 2023). HTTP filtering allows for precise filtering (keyword filtering), yet the widespread adoption of HTTPS has simplified the circumvention process (Hall et al., 2023). DNS-based censorship is akin to a more balanced censorship approach that is extensively utilized globally. Nevertheless, circumventing DNS-based censorship is relatively simple, particularly when it is executed at local DNS resolvers.

3. Related work

In this section, the status of Internet censorship in different countries is discussed, with emphasis placed on the techniques employed and the categories of censored content. When the topic of Internet censorship is broached, China frequently emerges as a primary example. China maintains the most extensive censorship apparatus globally, referred to as the Great Firewall of China (GFW). The GFW operates as a complex, multi-tiered firewall that manages traffic at the domain level, utilizing techniques such as DNS injections and filters applicable to both HTTP and HTTPS traffic. In late 2021, China introduced a novel censorship strategy that passively identifies and subsequently obstructs fully encrypted traffic in real-time, thereby catching researchers and circumvention tools off guard (Hoang et al., 2024). Recent research has identified emerging regional censorship systems, with Henan province deploying Transport Layer Security (TLS) / Server Name Indication (SNI) blocking (Wu et al., 2025). The research also indicated the presence of HTTP/URL filtering middle-boxes that inspect and block traffic outbound the province. The categories of censored content encompass file sharing, adult material, critiques of businesses and government.

Table 1
A comparison of censorship methods.

Aspect	TCP/IP blocking	HTTP/URL filtering	DNS-based censorship
Identification	"Destination IP Address" field	"Hostname" field	"Question" field
Interference	Packet dropping	1) TCP packet injection 2) Block-page injection	1) Erroneous response. 2) False IP address.
Advantages	1) Easy to implement. 2) Robust.	Used to obtain fine-grained filtering such as keyword/ URL filtering.	1) Easy to implement.
Dis-advantages	1) Over-blocking. 2) Cannot deal with services that are hosted on many IP addresses.	1) Over-blocking when used in keyword filtering. 2) Ineffective against HTTPS.	1) Easy to bypass when implemented at local DNS servers.
Visible effect	1) Connection timeout.	1) Connection reset. 2) Block-page	1) NXDOMAIN. 2) SERVFAIL. 3) REFUSED. 2) Block-page.

Iran, in turn, holds the position of having the second-highest level of Internet censorship globally (Master and Garman, 2023). Executing a thorough strategy aimed at regulating the flow of information and steering users towards a domestic national information network. The research by (Tai et al., 2025) suggested that the Great Firewall of Iran employs multiple blocking mechanisms to censor Internet traffic, DNS poisoning and HTTP blocking were among those mechanisms. While the authors in (Bock et al., 2020) stated that Iran uses a protocol filter as another layer of censorship, demonstrating an evolved censorship model that effectively isolates Iranian users from the global Internet while keeping domestic services intact. The range of content that was blocked includes adult content, followed by categories related to entertainment and gambling (Tai et al., 2025).

Similarly, Turkmenistan is characterized by one of the most restricted Internet landscapes globally. Since 2006, it has been identified as one of the rare Internet adversaries by Reporters Without Borders. The authors in (Nourin et al., 2023) revealed that Turkmenistan censors more than 122,000 domains, using different blacklists for each protocol (DNS, HTTP, and HTTPS). The research also identified 6000 over-blocking rules causing incidental filtering of more than 5.4 million domains. The measurement study used remote measurement techniques that exploit the bidirectionality of the firewall, which filters incoming packets as well as outgoing ones. The primary categories targeted by the filtering middle-boxes include adult content which constitutes nearly 25% of all blocking regulations.

As for Arab countries, Syria and Saudi Arabia are good examples. Syria's Internet censorship under the Assad regime (until December 2024) was characterized by comprehensive surveillance and targeted blocking. Analysis of leaked 600GB worth of logs from seven Blue Coat SG-9000 proxies deployed in Syria (Chaabane et al., 2014), uncovered a relatively discreet yet highly focused censorship mechanisms. Employing TCP/IP blocking and URL-based filtering to restrict access to certain subnets or websites, as well as utilizing keywords or categories to target types of content. This keyword-driven censorship resulted in unintended consequences, as numerous requests were denied even when they were unrelated to sensitive material. The primary focus of the blocked content was on social media platforms and instant messaging applications, including Facebook and Skype.

Saudi Arabia implements comprehensive Internet filtering through centralized infrastructure and sophisticated technical means. As reported by (Alharbi et al., 2020), the techniques employed encompass SNI blocking, HTTP-URL filtering, and DNS-based censorship. Saudi Arabia connects to the Internet via two national data service providers. The government frequently obstructs access to websites that promote violent extremist content, pornography, gambling, illegal drugs, and unauthorized use of copyrighted materials (Bishop et al., 2025).

Moving toward decentralized censorship implementations, Russia has intensified its Internet censorship dramatically following its invasion of Ukraine in 2022. Russian ISPs adopt multiple censorship techniques including DNS and keyword-based censorship (Ramesh et al., 2020). Recent research concerning Russia has indicated that the country intends to transition towards a more centralized approach to censorship. The previously decentralized model, where each ISP applies distinct blacklists, is being replaced by a more unified system. The Technical Means of Countering Threats system (referred to as TSPU) oversees the blocking registry in a more consistent and efficient manner, representing a model of decentralized deployment with centralized control (Xue et al., 2022). The targets of censorship have expanded significantly, and mostly focus on inbound traffic from foreign actors (Ramesh et al., 2023), as well as outbound censorship targeting western platforms such as twitter, BBC, Meduza, Deutsche Welle (Xue et al., 2022).

Another example of decentralized censorship is India; the country has a complex censorship landscape characterized by frequent Internet shutdowns and selective content blocking. Censorship techniques utilized in India differ among ISPs, who adopt various technical strategies.

Indian ISPs implement censorship methods based on DNS and HTTP protocols. The researchers in (Singh et al., 2020) found that some prominent ISPs cater to 49.7% of Internet users in India, implement SNI blocking. Additionally, various ISPs utilize a range of censorship methods, including HTTP/URL filtering and DNS-based censorship. The range of content that is subject to censorship goes beyond just pornography and gambling, encompassing political and social discourse as well (Yadav, 2018).

Table 2 summarizes prior studies and highlights key gaps addressed by this work. This study incorporates four local residential vantage points within Iraq, combines DNS and TCP/IP scanning techniques, and includes a unique test list provided by the MoC. It further examines over-blocking at the TCP/IP layer, where additional domain names are affected due to shared hosting.

4. Methodology

This section provides a discussion on how active real-time measurements have been utilized for assessing Internet censorship in Iraq. The tools and utilities utilized in the censorship detection process are also described in this section.

4.1. DNS-based censorship detection

Metrics show that Iraqi ISPs have minimal adoption of DNS encryption extensions like DNS over TLS (DoT) and DNS over HTTPS (DoH). Specifically, the usage of Domain Name System Security Extensions (DNSSEC) is minimal to non-existent within Iraq. Cloudflare's Radar statistics (Cloudflare 2026) reveal that barely a fraction of DNS queries to their resolver (1.1.1.1) from Iraq is DNSSEC aware, generally fluctuating between 0.1% and 0.2% of overall queries in 2025. Since most DNS requests are transmitted unencrypted, censors can identify which website a client is trying to resolve. Using either DNS servers they control, or packet injection from routers, censors can forge responses carrying DNS error codes such as "host not found" (NXDOMAIN), or provide non-routable IP addresses (Niaki et al., 2020).

Before explaining the DNS detection process, it is important to explore the DNS response codes, to better understand how Algorithm 1 works. Table 3 summarizes those response codes according to RFC1035 (Mockapetris, 2025). The "NOERROR" state signifies that the name server successfully interpreted the query. In contrast, "FORMERR" denotes the name server's failure to interpret the query due to its malformed nature or non-compliance with expected formatting standards. "SERVFAIL" indicates a temporary issue with the name server. Meanwhile, "NXDOMAIN" represents the absence of the domain name referenced in the query. The term "NOTIMP" implies that the name server does not accommodate the requested type of query. Lastly, in the case of "REFUSED", the name server declines to execute the specified operation due to policy considerations.

All the response codes mentioned above signify that the query has been successfully transmitted to the authoritative name server. However, there exists a scenario where no response is received for the query within the specified timeframe. This scenario does not represent a valid response code included in the DNS reply and has not been addressed in

Table 2
Summary of prior work on Internet censorship measurement.

Study	Country	Technique(s)				Vantage Points	Test List	Over-blocking analysis	Measurement Period	Limitation(s)
		TCP/IP	HTTP/URL	TLS/SNI	DNS					
(Hoang et al., 2024) (Hoang et al., 2024)	China		✓	✓	✓	Remote controlled machines	Public lists only	✓	Feb. 2022-Sep. 2023	▪ No residential probes.
(Wu et al., 2025) (Wu et al., 2025)	China		✓	✓		Remote VPSes.	Public lists only.		Nov. 2023-Mar. 2024 Oct. 2024-Mar. 2025	▪ No residential probes. ▪ No over-blocking analysis.
(Tai et al., 2025) (Tai et al., 2025)	Iran		✓		✓	Remote (non-responsive IP addresses)	Public lists only.		Nov. 2024-Jan. 2025	▪ No residential probes. ▪ No over-blocking analysis.
(Bock et al., 2020) (Bock et al., 2020)	Iran		✓	✓	✓	Local + remote	Public lists only.		Not specified	▪ No over-blocking analysis.
(Nourin et al., 2023) (Nourin et al., 2023)	Turkmenistan		✓	✓	✓	Remote VPSes.	Public lists only.	✓	Aug. 2022-Sep. 2022	▪ No residential probes.
(Alharbi et al., 2020) (Alharbi et al., 2020)	Saudi Arabia	✓	✓	✓	✓	Local residential	Public lists only.		Mar. 2018-Apr. 2020	▪ No over-blocking analysis.
(Ramesh et al., 2020) (Ramesh et al., 2020)	Russia	✓	✓		✓	Local + remote	Private lists only.		Not specified	▪ No over-blocking analysis.
(Ramesh et al., 2023) (Ramesh et al., 2023)	Russia	✓	✓	✓	✓	Local + remote	Public lists only.		Mar. 2022-May. 2022	▪ No over-blocking analysis.
(Xue et al., 2022) (Xue et al., 2022)	Russia	✓		✓		Local + remote	Public + Private lists.		Feb. 2022-Mar. 2022	▪ No over-blocking analysis.
(Singh et al., 2020) (Singh et al., 2020)	India	✓	✓	✓	✓	Local residential	Private lists only.		Not specified	▪ No over-blocking analysis.
This work	Iraq	✓			✓	Local residential	Public + Private lists.	✓	May. 2025-Jun. 2025 Aug. 2025-Sep. 2025	▪ No HTTP/TLS-based tests.

Algorithm 1
DNS-based censorship detection.

Algorithm 1: DNS-based censorship detection

```
Input: list_of_domains
Output: censored_domains
Procedure QUERY_DNS(domain, resolver)
    DNS response ← query domain via resolver
    return DNS response
Procedure IS_CENSORED(domain) // Analyze response
    response_isp ← QUERY_DNS(domain, isp_resolver)
    status_isp ← response_isp.status
    if status_isp = TIMEOUT then
        return (True, TIMEOUT)
    if status_isp ∈ {NXDOMAIN, SERVFAIL, REFUSED} then
        response_ref ← QUERY_DNS(domain, reference_resolver)
        if response_ref.status ≠ status_isp then // Validate using reference resolver
            return (True, status_isp)
    if status_isp = NOERROR AND response_isp.answers = ∅ then
        return (True, EMPTY_RESPONSE)
    return (False, None)
    censored_domains ← ∅
    for each domain ∈ list_of_domains do
        (flag, reason) ← IS_CENSORED(domain)
        if flag = True then
            censored_domains ← censored_domains ∪ {domain}
    return censored_domains
```

Table 3
DNS response codes.

Response Code	DNS Response Status	Description
0	NOERROR	No error condition
1	FORMERR	Format error.
2	SERVFAIL	Server failure.
3	NXDOMAIN	Name error.
4	NOTIMP	Not implemented.
5	REFUSED	Refused.

RFC1035. Rather, it is utilized by DNS tools (such as nslookup, ZDNS, MassDNS, etc.) to denote that no response was received for the query within the allotted time. These tools refer to this situation as “TIMEOUT”.

Algorithm 1 is employed for DNS-based censorship detection, through a cross-reference of responses from an ISP-assigned resolver and a reference resolver (which in this case is Google DNS). Other public DNS resolvers—such as Cloudflare (1.1.1.1) and Quad9 (9.9.9.9)—could be incorporated in future work to improve the robustness and accuracy of the reference resolution process. Notably, many domains—especially those fronted by Content Delivery Networks (CDNs)—use GeoDNS and other traffic-steering mechanisms that can legitimately return different A/AAAA records depending on the location of the recursive resolver and the client network. Therefore, our DNS-based inference does not treat “different IP answers” across resolvers as censorship evidence. Instead, Algorithm 1 relies primarily on response failure/denial patterns (e.g., NXDOMAIN, SERVFAIL, REFUSED, TIMEOUT, or NOERROR with empty answers) observed at the ISP resolver and then cross-checks whether public resolvers return resolvable outcomes. Consequently, our reference resolver should be interpreted as a reference vantage for resolvability verification rather than as an oracle for the exact record set expected in Iraq. Nevertheless, resolver-specific behavior and transient authoritative failures remain possible confounders; consequently, isolated DNS anomalies are interpreted conservatively, with emphasis placed on repeated/ consistent patterns observed over time.

Since the focus of our approach is on ISP-assigned resolvers, alternative measurement approaches have been proposed that target non-responsive IP addresses or dedicated control servers. These approaches are primarily designed to detect DNS injection by intermediary middleboxes. In such cases, DNS queries are directed to servers that are not expected to generate legitimate responses, often requesting the resolution of test domains. The absence of a response is interpreted as an indication that no packet injection has occurred along the network path, thereby suggesting that the queried domain is not subject to censorship. Conversely, the presence of any response is taken to indicate that DNS injection has been performed by an intermediate device.

The procedure “QUERY_DNS” is defined to execute DNS lookups, taking a domain name and a resolver as input. A second procedure, “IS_CENSORED”, is defined to accept a domain input and perform response analysis. When the response status is “TIMEOUT”, censorship is inferred, as this indicates that DNS queries are being dropped along the path. When the response status is “NXDOMAIN”, “SERVFAIL”, or “REFUSED” the response is verified by repeating the measurement with a reference resolver. If the reference resolver returns a different status (e.g., “NOERROR”), censorship is inferred; otherwise, the domain is classified as not censored. For responses with status “NOERROR”, the content of the response is examined; an empty response is interpreted as censorship, whereas a non-empty response is not. This procedure is applied iteratively over all domains in the input lists, (the “Majestic Million” domains list, and the private domains list), analyzing the status of each domain. DNS measurements have been carried out utilizing ZDNS (Izhikevich et al., 2022), an open-source tool capable of generating a substantial volume of DNS queries within a brief timeframe. ZDNS has been extensively referenced in the literature (Ramesh et al., 2020, Williams et al., 2024, Xu et al., 2023).

4.2. TCP/IP blocking detection

Censors can block access to websites by preventing clients from connecting to the specific IP addresses the website is hosted on. This method has been specifically used to block circumvention proxies, and is how China prevents access to the Tor network (Sundara Raman et al., 2020).

The methodology of this study adheres to a straightforward principle: sending probing packets aimed at the IP addresses of each domain from the two input lists (the “Majestic Million” domains list, and the private domains list). Subsequently, hosts that do not reply to probing packets from test networks but do respond appropriately from reference control networks are identified as censored. The test networks are the networks to be evaluated, particularly ISP1, ISP2, ISP3, and ISP4 (discussed in Section 5.5). As the reference control network, Google Colab was employed to validate instances of TCP/IP blocking. Because this network is located outside Iraq, traffic appears to be proxied through another country, thereby enabling cross-referencing between local measurements and an external baseline.

To execute this methodology, Nmap was employed, a widely recognized network scanning tool designed for network discovery and security auditing. Nmap was selected on the basis of its exceptional accuracy, which allows for precise management of scan timing and rate limiting, and its versatility that accommodates various scan types and offers extensive packet customization options. Nmap offers a multitude of configuration options and scanning techniques, facilitating the creation of virtually any packet sequence (Al-Khazaali et al., 2025).

A dual scanning approach was employed, comprising an Internet Control Message Protocol (ICMP) ping scan and a TCP SYN ping scan. The ICMP ping scan serves as a basic layer 4 discovery scan that identifies active hosts by sending straightforward ICMP type 8 code 0 "Echo Request" packets, with the anticipated response being an ICMP type 0 code 0 "Echo Reply" packet.

The TCP SYN ping scan utilizes TCP packets that have SYN flags activated, targeting ports 80 (typically used for HTTP) and 443 (generally used for HTTPS). The anticipated reply to a TCP SYN packet is either a TCP SYN/ACK packet or a TCP RST packet. The TCP SYN ping is regarded as stealthy since it avoids the completion of full TCP handshakes. Rather, it replies to TCP SYN/ACK packets with TCP RST packets, promptly terminating connections and minimizing firewall disruptions linked to the establishment of complete TCP connections across various ports (Lyon, 2008).

Fig. 1 depicts three scenarios in which hosts are deemed active: (a) hosts reply to TCP SYN packets with TCP SYN/ACK packets, signifying that the probed port is open; (b) hosts reply to TCP SYN packets with TCP RST packets, indicating that the probed port is closed while the host remains accessible; and (c) hosts respond to ICMP Echo requests with ICMP Echo replies, indicating host responsiveness. The utilization of TCP packets addresses situations where hosts may discard ICMP packets, as firewalls often obstruct ICMP traffic to mitigate ping flood attacks. Our empirical observations revealed that certain hosts display contrary behavior, responding solely to ICMP packets even when they originate from the same source IP addresses.

Algorithm 2 outlines the methodology for TCP/IP censorship detection. This algorithm necessitates two inputs: a list of domain inputs and the data pertaining to hosted domains from IPinfo. The procedure "MAPPING" is introduced, which takes a domain input and conducts domain-to-IP lookups within the hosted domains data provided by IPinfo, to obtain the relevant IP addresses. After that, the procedure "HOST_SCAN" is defined, which accepts an input IP address and a network vantage point. This procedure handles TCP SYN ping scan and ICMP ping scan mentioned above (Section 4.2). Subsequently, all domains are traversed and subjected to domain-to-IP lookups via the "MAPPING" procedure. The resultant IP addresses are added to a list named "Test_list". Each host in the "Test_list" is scanned within a residential network using the "HOST_SCAN" procedure. Hosts that respond are included in the "Hosts_reachable" list. Conversely, non-responsive hosts are evaluated in the same manner within the reference network (Google Colab). Hosts that remain unresponsive are recorded in the "Hosts_unavailable" list, while those that respond in the reference network are identified as censored and added to the "Censored_IP_Addresses" list.

In addition to this approach, a mechanism known as censorship device location identification leverages traceroute techniques to infer the position of filtering devices within a network. This method utilizes ICMP packets with incrementally varied Time-to-Live (TTL) values to identify the last network device that processes each packet before it is dropped or altered.

5. System design

The system design resembles a single-end measurement framework, as measurements are conducted from the client perspective only. The high-level idea is that measurement devices are configured to issue probing packets that embed test domain names/IP addresses, thus actively triggering censorship devices. The assessment starts with DNS censorship measurements, followed by TCP/IP measurements. The measurement setup and vantage points used in conducting the measurements are also explained in this section.

5.1. DNS-based censorship detection pipeline

The first measurement is a DNS measurement, which is conducted using the default ISP resolvers. First, the list of domains is prepared, including the "Majestic Million" domains list and the private domains list, with one domain per line. Next, measurements are initiated in parallel for all domains in the input list. The raw DNS responses are then collected in a JavaScript Object Notation (JSON) file. Subsequently, all responses are analyzed based on the response code and answer fields (as described in Section 4.1), and censorship is validated using the reference resolver. Domain names that are classified as non-censored are resolvable, while the remaining domains are classified as censored via DNS

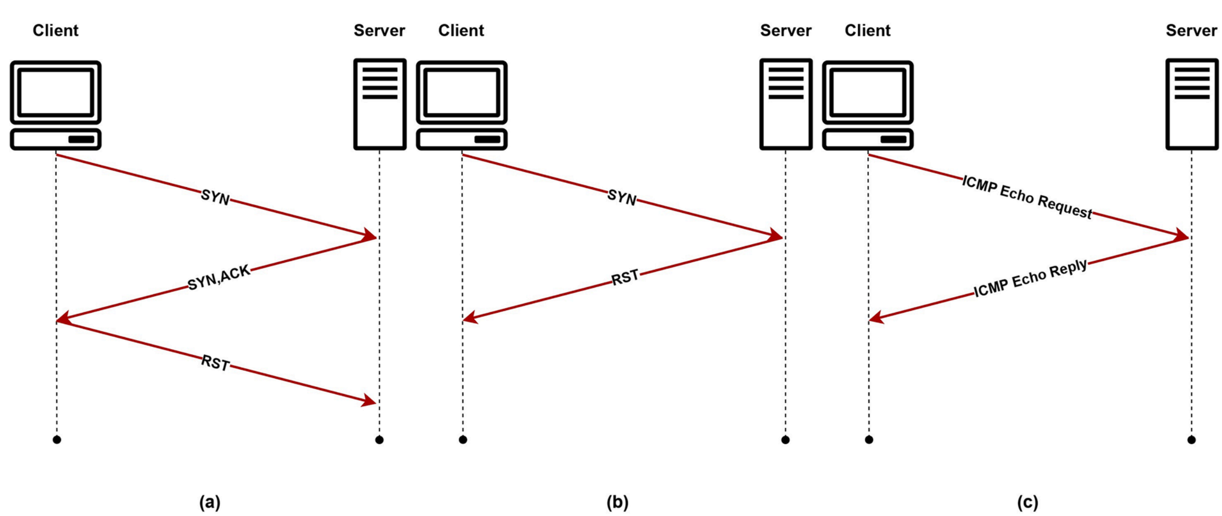

Fig. 1. Scenarios indicating host availability as determined by Nmap: (a) TCP SYN ping to an open port; (b) TCP SYN ping to a closed port; (c) ICMP ping.

Algorithm 2
TCP/IP blocking detection.

```
Algorithm 2: TCP/IP blocking detection
Input: domain_input_list, IPinfo_data
Output: Censored_IP_Addresses
Procedure MAPPING(domain, IPinfo_data) // Domain to IPs mapping
    return { ip | entry ∈ IPinfo_data AND domain ∈ entry.domains }
Procedure HOST_SCAN(ip, vantage_point) // Perform host scan
    Perform TCP SYN probe on ports {80, 443}
    Perform ICMP probe
    if any probe receives a response then
        return True
    else
        return False
Hosts_reachable ← ∅, Hosts_unavailable ← ∅, Censored_IP_Addresses ← ∅
unique_ips ← ∅
for each domain ∈ domain_input_list do // Construct test list
    ips ← MAPPING(domain, IPinfo_data)
    unique_ips ← unique_ips ∪ ips
Test_list ← unique_ips
for each host ∈ Test_list do
    result_res ← HOST_SCAN(host, residential)
    if result_res = False then
        result_ref ← HOST_SCAN(host, reference_network)
        if result_ref = True then
            Censored_IP_Addresses ← Censored_IP_Addresses ∪ {host}
        else
            Hosts_unavailable ← Hosts_unavailable ∪ {host}
    else
        Hosts_reachable ← Hosts_reachable ∪ {host}
return Censored_IP_Addresses
```

censorship. The proposed method may undercount censorship in cases where the ISP resolver returns an alternative but functional IP address instead of failing outright. For example, a censored domain may resolve to a sinkhole, warning page, or other substituted destination that appears valid at the DNS level. Because the current approach focuses on discrepancies between failed and successful resolution, such cases may remain undetected. Future work could address this limitation by validating returned IP addresses against external IP intelligence sources and by inspecting HTTP content to identify substituted or blocked destinations. These extensions would likely identify additional censorship instances. Fig. 2 shows the proposed design for DNS-based censorship, a visual representation of Algorithm 1.

5.2. TCP/IP blocking detection pipeline

The TCP/IP measurement framework processes IP addresses rather than domains as input to prevent DNS load balancing from affecting continuous measurements. The first step is domain-to-IP mapping,

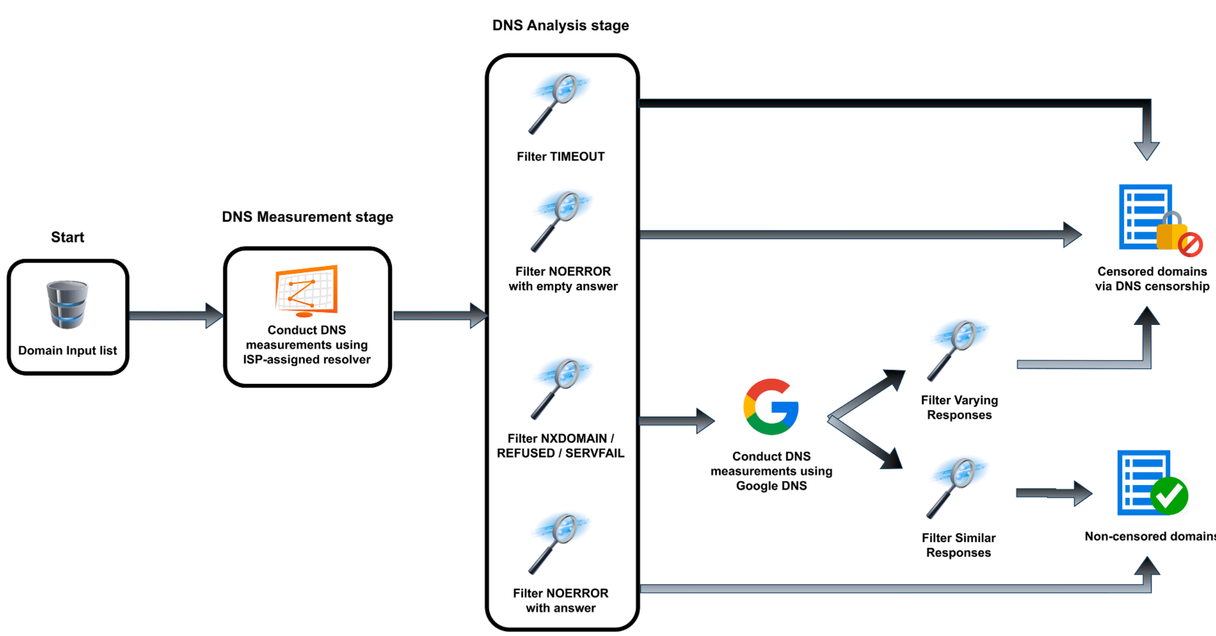

Fig. 2. DNS-based censorship detection pipeline.

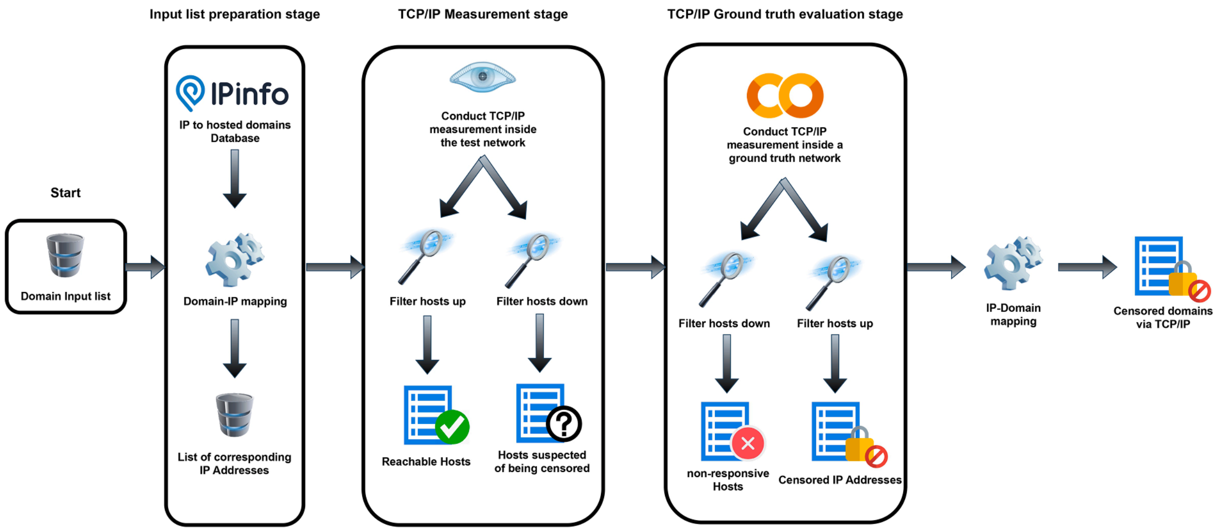

Fig. 3. TCP/IP blocking detection pipeline.

accomplished using the "hosted domains" database provided by IPinfo (IPinfo 2025). As a prominent provider of IP intelligence, IPinfo upholds extensive databases that encompass IP geolocation, company mapping, mobile carrier identification, AS mapping, and hosted domain relationships. IPinfo's hosted domains data offers bidirectional utility: IP-to-hosted domains and domain-to-IP mappings. The primary mapping identifies all hosted domains on individual IP addresses. Conversely, domain-to-IP mapping extracts all IP addresses hosting specific domains, proving valuable for establishing domain-IP relationships. Subsequently, TCP/IP measurements are conducted as described in Section 4.2 for all IP addresses in parallel. Responsive hosts—those successfully answering probing packets—are considered reachable within test networks and added to the "Reachable Hosts" list. Any host that is online in at least one measurement is considered accessible, thereby reachable within test networks. Non-responsive hosts represent the primary focus for censorship analysis. A host must be offline in all measurements to be considered non-responsive. Those non-responsive hosts undergo re-evaluation using networks outside the target jurisdictions, which serve as reference networks. Google Colab is used as a reference network for measurement validation. Hosts that remain unresponsive in the reference network likely indicate temporary unavailability due to server overload, maintenance, or permanent unavailability. These hosts are added to the "Host unavailable" list. Hosts that respond from the reference network but remain unresponsive from test networks are classified as censored and added to the "Censored IP Addresses" list. Finally, censored IP addresses are mapped to their corresponding domains using IPinfo's hosted domains data. Domains are classified as censored only when all associated IP addresses for that domain are censored. This reduces false positives caused by depending on single IP addresses. As discussed in Section 4.2, a dual scanning approach was employed; ICMP ping scan and a TCP SYN ping scan to reduce firewall interference on specific protocols (Lyon, 2008). Furthermore, Nmap default scanning parameters have been chosen, so that scan traffic looks normal for ISPs (Al-Khazaali et al., 2025). Fig. 3 illustrates the detection process, a visual representation of Algorithm 2.

A limitation of any reachability-based censorship methodology is that modern hosting infrastructures (in particular CDNs) may exhibit geo-dependent behavior. The same hostname may map to different CDN edges across networks, and anycasted IP addresses can route clients in different regions to different Points-of-Presence (PoPs). As a result, "reachable from Google Colab but not reachable from Iraq" can, in some cases, also arise from CDN steering, regional PoP incidents, or routing/peering differences rather than deliberate ISP-level blocking. To reduce sensitivity to localized failures, our study performs measurements from four independent residential vantage points spanning distinct market segments, focusing on persistent non-responsiveness that is consistent across vantage points rather than single-probe outcomes.

In light of the above, the obtained TCP/IP results should be interpreted as evidence of observed access restriction from Iraqi residential networks. While consistent in-country failures combined with external reachability are strongly suggestive of filtering, definitive attribution (e.g., distinguishing ISP filtering from remote geo-restrictions or CDN policy decisions) may require additional application-layer validation (e.g., TLS SNI/HTTP Host-based checks) that is beyond the scope of this measurement pipeline and left for future work. Furthermore, the incorporation of IP mapping datasets, such as IPinfo's hosted domains dataset, into the pipelines of censorship measurement systems can be generalized to improve the effectiveness of TCP/IP blocking detection. Additionally, this dataset plays a key role in enabling the analysis of TCP/IP over-blocking.

5.3. *Test lists*

Due to the discontinuation of the widely-used Alexa ranking service, we turned to the top 50,000 domains from the "Majestic Million" dataset (Majestic 2025). This dataset comprises one million domains that have the highest number of referring IP addresses. The FortiGuard Labs Web Filter classification system (Fortinet 2025) has been leveraged to classify each domain in this test list. Fig. 4 shows the classification of the Majestic test list, illustrating the top 45 categories out of 90 categories. We chose the top 50,000 domains because they constitute a statistically significant segment of the dataset, as determined by prior research (Rweyemamu et al., 2019). The remaining entries may be affected by ranking inaccuracies, randomness, and possible non-existence. Furthermore, a private test list obtained exclusively from the Iraqi MoC has been included, which comprises over 100,000 domains. This test list has been classified using the same FortiGuard classification system. The majority of domains in this list are classified as "pornographic", as revealed in Fig. 5. A logarithmic scale was employed in Fig. 5 to clearly represent variations among different categories, as this was not clear when using the linear scale. Although public test lists are essential for censorship analysis and ranking patterns, the incorporation of a private test list can enhance censorship measurement pipelines by uncovering previously unobserved censorship patterns and reducing potential bias in the results.

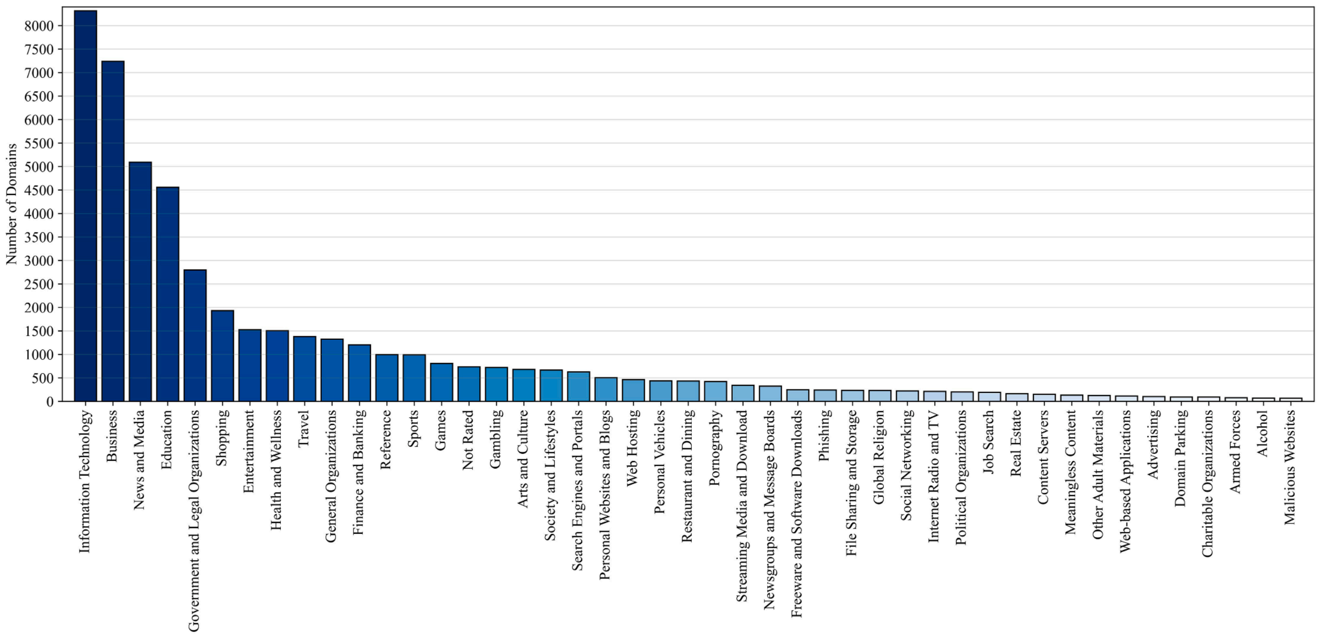

Fig. 4. Classification of Majestic test list (top 45 categories out of 90).

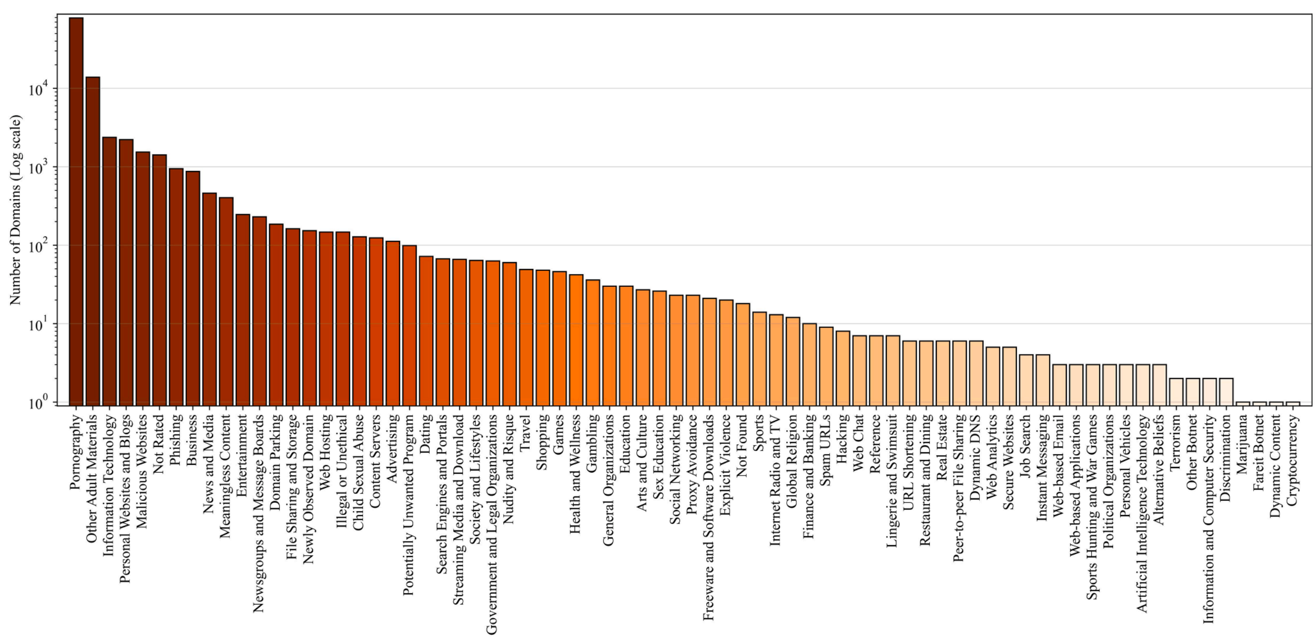

Fig. 5. Classification of the private test list (log scale).

5.4. Measurement setup

The Majestic test list was evaluated from May 30 to June 29, 2025, whereas the private test list was assessed from August 20 to September 20, 2025. Active measurements were conducted three times a day (at 00:00, 08:00, and 16:00 UTC+03). These intervals were selected to coincide with off-peak hours, thus improving measurement efficiency. This timing facilitated the analysis of shutdown impacts on measurements from various perspectives. Both measurement periods experienced considerable Internet outages, which, as noted by (Al-Musawi et al., 2025), happen during baccalaureate examination periods. During the first measurement period, 13 scheduled measurements were not completed, leading to 78 successful measurements out of a planned total of 91 (covering the timeframe from May 30, 2025, at 4:00 PM to June 29, 2025, at 4:00 PM). In the second measurement period, 14 scheduled measurements were also unfulfilled, resulting in 80 successful measurements from a planned total of 94 (spanning from August 20, 2025, at 4:00 PM to September 20, 2025, at 4:00 PM). Other than national Internet shutdowns mentioned above, no notable ISP outages or power disruptions were observed. For the final analysis, only completed measurements were included, while partial measurements recorded during these shutdown periods were excluded. DNS measurement campaigns took approximately 5–15 min to complete, depending on the test list and resolver state, reflecting the stateless nature of the DNS protocol and the efficiency of the tool. By contrast, TCP/IP measurement campaigns took approximately 30–90 min because of Nmap's default probing behavior, with duration varying by test list and ISP state.

5.5. Vantage points

To determine whether censorship behavior remains consistent across ISPs—thereby assessing centralized versus decentralized censorship implementation— measurements were performed across four distinct residential networks. In this study, each service provider is labeled anonymously (e.g., ISP1, ISP2, …) instead of using real names, to follow ethical practice, protect reputations, and avoid unintended harm or misinterpretation to the broader public.

Each of these networks exemplifies a distinct market segment; for example, ISP1 is among the most recent providers in southern Iraq. ISP2 stands as one of the largest ISPs in Iraq. Furthermore, we selected the premier ISP that caters to Baghdad, the capital of Iraq, denoted as ISP3. In addition, we incorporated one of the largest mobile network operators in the nation, which we designate as ISP4. This selection enables comprehensive analysis of censorship implementation across major ISPs, dominant mobile carriers, prominent regional operators, and emerging market entrants.

Measurement automation across all networks utilized four Raspberry Pi devices: specifically, Raspberry Pi 3 Model B version 1.2 units equipped with 1 GB RAM, ARMv7 architecture, and running Raspbian GNU/Linux version 10 (Buster). These devices connected to each test network to conduct censorship measurements. All Raspberry Pi devices are connected to an Application Programming Interface (API) endpoint. The endpoint stores all measurement files from all ISPs with their associated timestamps, enabling easier accessibility. A central analysis machine is used to process all measurement files from all vantage points, see Fig. 6.

The central analysis machine runs Windows 10 and is equipped with an Intel® Core™ i7–11800H CPU (2.30 GHz) and 16 GB of RAM. Its role extends beyond data storage. Specifically, the machine collects raw measurement files from all the residential vantage points via the API, ensuring that all necessary files are stored centrally for analysis. It automatically executes pipelines, including domain-to-IP mapping, domain classification, result processing, and preparation of processed datasets for visualization. The analysis is automated utilizing Python v3.12.6, which is also used for visualization. Accordingly, Raspberry Pi devices are used only for conducting measurements at the specified time intervals and to upload the resulting raw measurement files, together with their timestamps, to the API.

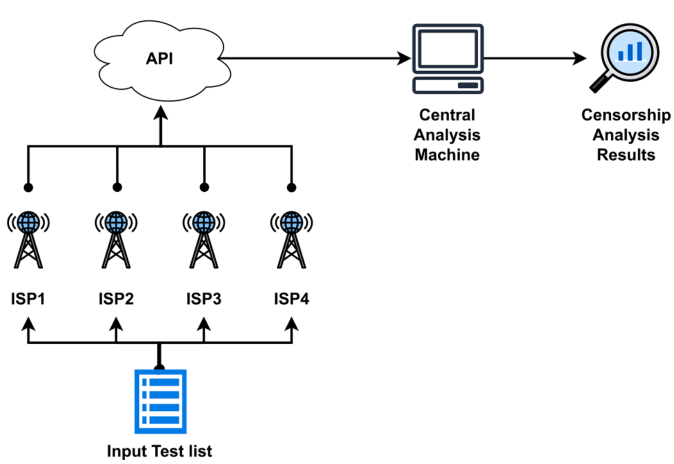

Fig. 6. Data collection process.

5.6. Ethical considerations

Although large-scale network scanning is common on today's Internet, the sensitive nature of censorship measurement requires careful consideration of ethical and operational risks. Therefore, the study was designed to minimize potential legal exposure and unnecessary impact on networks and services. All volunteers involved in the study were also protected from avoidable legal risk. To preserve privacy, each ISP is anonymized using a unique label. The purpose of the study is solely scientific analysis and documentation, not to harm or affect the reputation of any ISP. In addition, the default parameters of the ZDNS and Nmap tools were retained to avoid overwhelming any network or server.

6. Results

As outlined in Section 5.3, measurements are conducted utilizing two distinct test lists, resulting in two separate scenarios. The first scenario involves the analysis of the most frequented websites, as indicated by the "Majestic Million" test list (Majestic 2025). The subsequent scenario pertains to a private test list acquired from the Iraqi MoC. For DNS-based analysis, the results are presented as confidence intervals (CI). The CI for a sample mean represents a set of values derived from sample data, which is expected to encompass the actual population mean with a predetermined level of confidence. This interval illustrates the degree of uncertainty associated with the sample mean as a representation of the population mean (Moore et al., 2013).

The interval is usually expressed as:

$$\text{CI} = \text{sample mean } (\mathscr{X}) \pm \text{margin of error} \tag{1}$$

$$\text{Margin of error} = \text{critical value } (z) * \text{Standard Error(SE)} \tag{2}$$

$$\text{For a 95\% confidence level}: \; z = 1.96$$

$$\text{Standard Error (SE)} = \text{population standard deviation } (\sigma) \, / \sqrt{\text{sample size (n)}} \tag{3}$$

The margin of error reflects the variability of the sample mean and is influenced by the sample size, the population standard deviation, and the selected confidence level (for instance, 95%). To illustrate, a 95% CI indicates that repeated sampling would yield intervals in which approximately 95% are expected to contain the true population mean.

6.1. Majestic test list scenario

The Majestic test list constitutes an open-source dataset comprising one million domains, curated on the basis of link intelligence data. The ranking methodology employed within this dataset prioritizes domains with the highest number of referring IP addresses. The dataset spans multiple content categories (see Fig. 4 for classification). Measurements utilizing the Majestic test list were conducted between May 30 and June 29, 2025. The analysis commences with the examination of DNS-based censored domains across ISPs, including the calculation of CI and an exploration of the correlation between domain ranking and DNS-based censorship. Subsequently, the study addresses TCP/IP blocking, identifying censored domains and associated IP addresses, and assessing the broader implications of TCP/IP blocking on other domains.

An analysis of all gathered DNS responses was performed utilizing Algorithm 1, and the findings were largely consistent across all measurements. Nevertheless, certain variations were observed, which are attributed to anycast, load balancing, and geo-routing. To address these inconsistencies, the results are presented as CI (discussed in Section 6). The CI offers an estimation of the true population mean; therefore, rather than computing the average number of censored websites, a range is determined that encompasses the majority of values. Fig. 7 illustrates

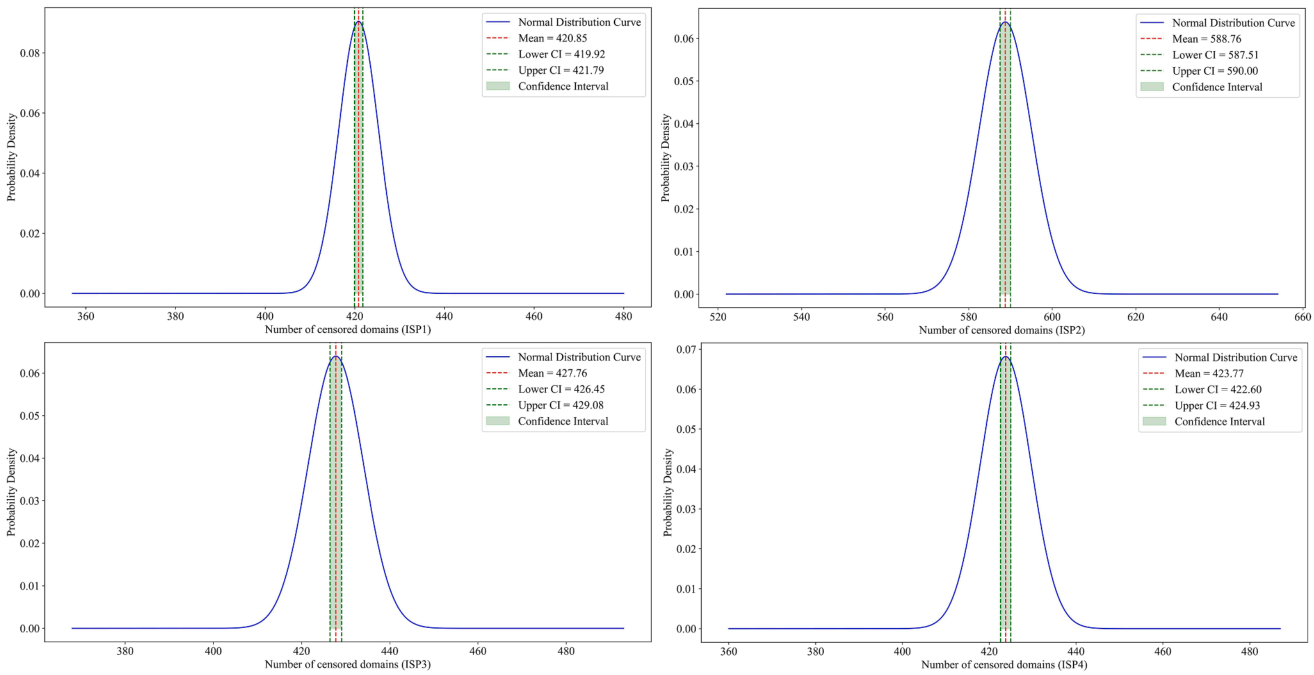

Fig. 7. Number of DNS-based censored domains expressed as a CI (Majestic test list scenario).

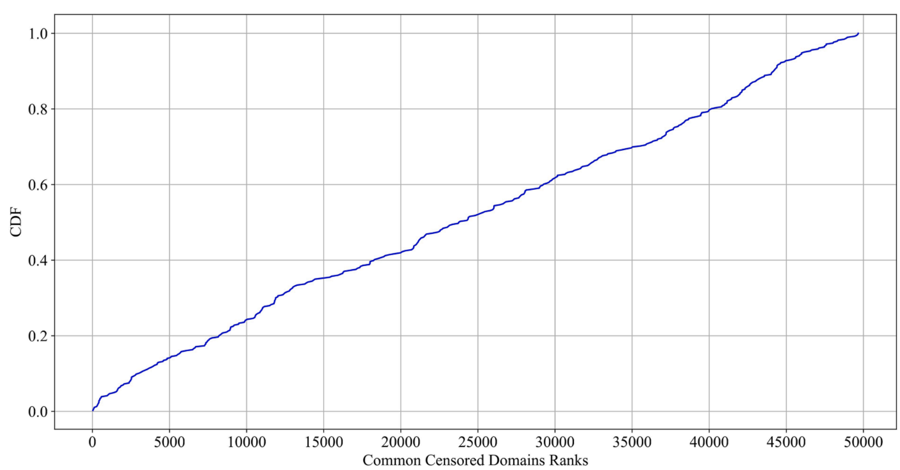

Fig. 8. CDF of common DNS-based censored domain rank.

the CI of DNS-based censored domains alongside the Gaussian distribution for each ISP, indicating that 95% of the observed values fall between the lower and upper limits. It is evident that the figures are relatively similar for each ISP, suggesting that this form of censorship is consistent across all ISPs.

Regarding the rankings of censored domains, an analysis was performed focusing on the instances where the lowest number of censored domains was observed for each ISP. Subsequently, the domains commonly censored across all ISPs were identified. This process resulted in a total of 386 domains that are subject to censorship by all ISPs. Fig. 8 illustrates the cumulative distribution function (CDF) of the ranks of these censored domains. It is evident that approximately 60% of the commonly censored domains fall within the top 30,000 domains, while around 20% are found within the top 10,000 domains. Furthermore, our analysis revealed that 16 domains are ranked among the top 1000 domains. This finding suggests that censorship is applied to domains irrespective of their rank and level of popularity.

For TCP/IP measurements, we identified hosts that consistently exhibited downtime across all measurements, which included the reference data collected from Google Colab. As a result, any host observed to be online in at least one measurement was classified as accessible. Fig. 9.a illustrates the shared censored IP addresses among the four networks. The vertical bars represent the number of censored IP addresses that are common to the networks, while the horizontal bars depict the total number of censored IP addresses for each individual network. Each vertical bar is associated with a specific combination of ISPs, denoted by the connected black dots beneath the bar; these combinations are referred to as "sets".

For illustrative purposes, we examine ISP1 (which is one of most recent providers in the south of Iraq) as a case study. ISP1 recorded a total of 203 censored IP addresses, which were divided into 5 distinct sets. The first set comprised the commonly censored IP addresses that were found across all ISPs, totaling 195 IP addresses. The second set included censored IP addresses that were unique to ISP1 and ISP4,

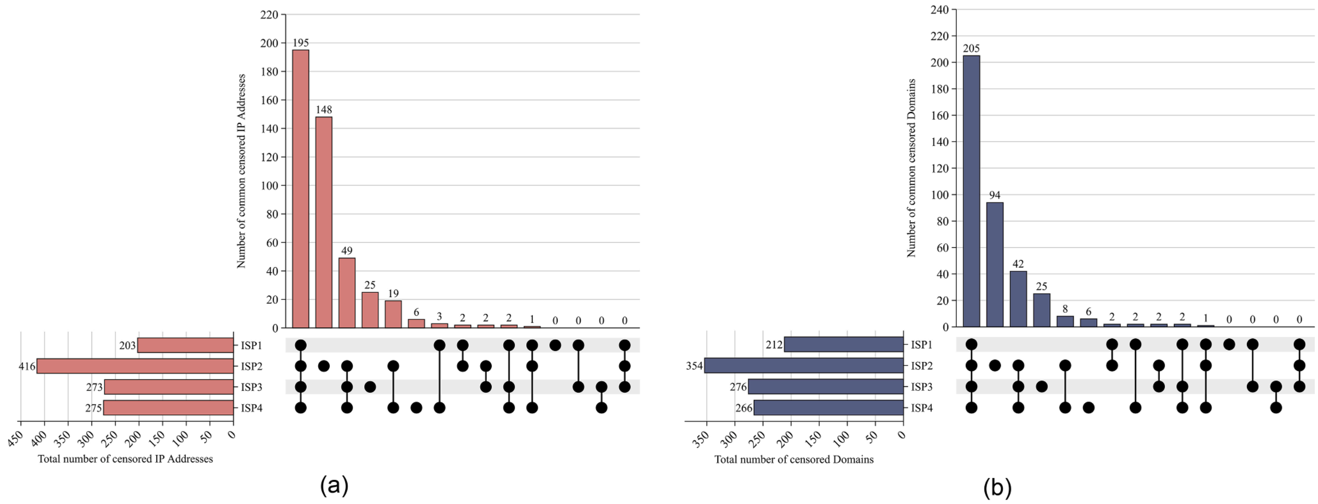

Fig. 9. Overlapping between ISPs (Majestic test list scenario): (a) IP-blocked addresses; (b) Domains affected by IP level blocking.

amounting to 3 IP addresses. The third set contained censored IP addresses specific to ISP1 and ISP2, with a total of 2 IP addresses. The fourth set encompassed censored IP addresses identified in ISP1, ISP3, and ISP4, also totaling 2 IP addresses. Lastly, the fifth set included censored IP addresses that were present in ISP1, ISP2, and ISP4, comprising 1 IP address. It is noteworthy that most of the censored IP addresses were common across all ISPs. ISP2 exhibited the highest number of unique IP addresses (those censored solely in ISP2), accounting for 148 IP addresses, while ISP1 had no unique IP addresses, as all of its censored IP addresses were shared with at least one other network.

A comparable trend was observed for censored domains in Fig. 9.b, where 205 domains were common across all ISPs, indicating the largest share. ISP2 had the highest count of unique domains—domains that were exclusively censored by this ISP—with 94 unique domains, while ISP3 and ISP4 followed with 25 and 6 domains, respectively. In contrast, ISP1 reported no unique domains. This implies that ISP1 does not enforce independent censorship policies but instead adheres to regulations set forth by other ISPs or higher authorities.

From Figs. 9a and b, it is evident that there are significant discrepancies among ISPs; certain domains were reachable from one network while simultaneously being unreachable from another. This phenomenon is attributed to the distinct policies implemented by each ISP. Older ISPs and major providers in Baghdad tend to impose the strictest filtering policies, likely due to closer government oversight and newer compliance frameworks.

According to (Hall et al., 2023), when an IP address is blocked, all domains associated with that IP address will be affected. This phenomenon is commonly known as over-blocking. To assess the extent of over-blocking, the hosted domains data from IPinfo was employed to identify all domains linked to each specific IP address. Subsequently, each domain was categorized through the classification services offered by FortiGuard Labs (Fortinet 2025). Fig. 10 illustrates the top five categories of affected hosted domains for each ISP. In addition to the "Pornography" category, the categories "Business" and "Information Technology" also rank among the top five categories across all ISPs, indicating that numerous domains are likely impacted by this type of censorship. Popular domains (such as archive.org) are also impacted by

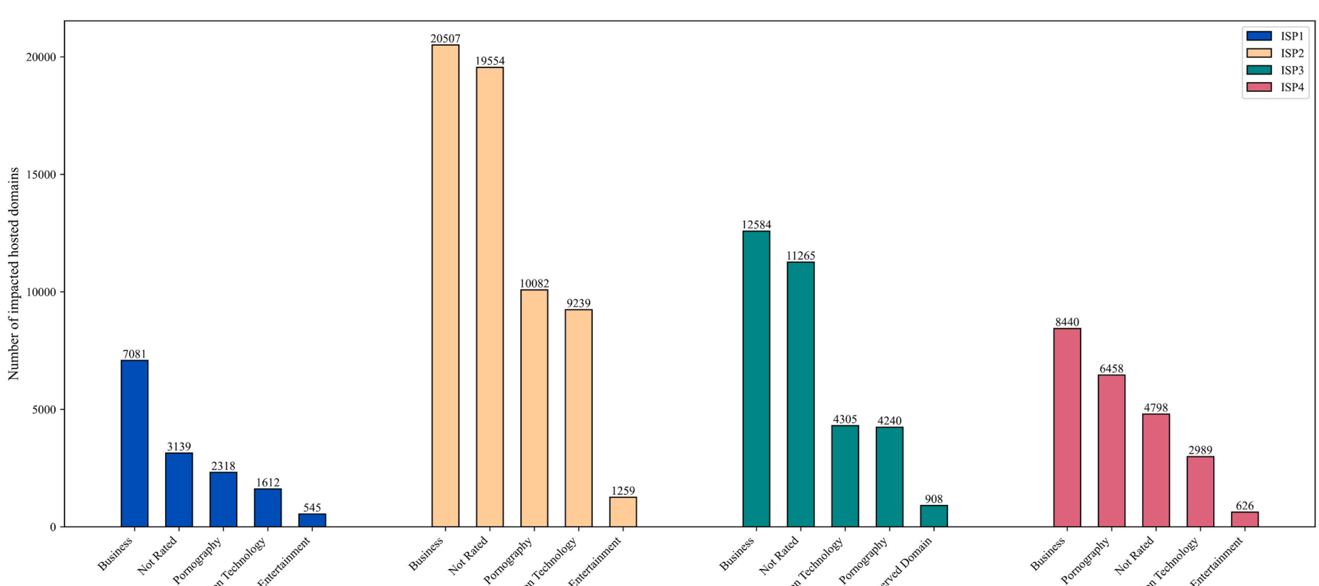

Fig. 10. Top 5 categories of impacted hosted domains for each ISP (Majestic test list scenario).

this TCP/IP blocking, due to the popularity of virtual hosting. Furthermore, a substantial number of educational domain names were found to share the same issue. The number of impacted hosted domains categorized as "Education" is 437, 727, 563, 456 for ISP1, ISP2, ISP3, and ISP4, respectively.

6.2. Private test list scenario

The private test list, obtained exclusively from the Iraqi MOC, encompasses over 100,000 domains (see Fig. 5 for categorization). Measurements based on this dataset were conducted between August 20 and September 20, 2025. The analytical procedure parallels that of the Majestic test list, commencing with the identification of DNS-based censored domains for each ISP, the calculation of CI, and a detailed account of the most frequently observed DNS responses per ISP. The subsequent stage of analysis highlights TCP/IP-level censorship, documenting censored domains and IP addresses, while also examining the collateral effects of TCP/IP blocking on other domains.

The quantity of DNS-based censored domains is significantly greater than that found in the Majestic test list. Most of these domains are categorized under adult-related categories (refer to Fig. 5). Fig. 11 shows the CI of DNS-based censored domains alongside the Gaussian distribution for each ISP. The pattern is relatively similar for ISP1, ISP2, and ISP3. Notably, for ISP4, the CI is considerably larger compared to the other ISPs; this can be attributed to the substantial fluctuations observed for this ISP throughout the measurement period, even though the same measurement parameters were employed. Overall, it can be concluded that the DNS-based censorship implemented exhibits similar behavior across all ISPs.

Fig. 12 illustrates the DNS responses observed for each ISP. The mean and standard deviation are calculated for each DNS response to evaluate the most frequently observed responses over time. It is evident that ISP1 predominantly returns "NXDOMAIN" and "TIMEOUT" for the majority of queries, suggesting the presence of DNS-based censorship. In the case of ISP2, 'NXDOMAIN' emerges as the prevailing response. In contrast, ISP3 predominantly yields empty responses, which lack answer fields and are marked with the return code "NOERROR". This situation diverges from the typical "NOERROR" scenario, where responses include valid answer fields. Furthermore, all empty responses are associated with the same authority, further indicating obvious DNS-based censorship. ISP4 employs "NXDOMAIN" as a means of censorship, although it is noteworthy that numerous domains are resolvable with the "NOERROR" state. This diversity in responses reflects the varying configurations of each ISP's resolver.

Moving toward TCP/IP blocking, the shared censored IP addresses among the four networks are shown in Fig. 13.a. Like Fig. 9.a, the horizontal bars show the total number of censored IP addresses for each network, while the vertical bars show the number of censored IP addresses shared by the networks. Clearly, the most common case is the leftmost set (censored IP addresses across all ISPs), which suggests a high degree of compliance. Nevertheless, a noticeable number of censored IP addresses are censored only in ISP2 and ISP3. This suggests that certain ISPs have a tendency to implement their own policies in spite of the high degree of compliance. As for the TCP/IP censored domains, the number is significantly larger than in the Majestic test list scenario, the majority are shared among all ISPs. Still, ISP2 and ISP3 have their own censored domains, which are not censored in other ISPs.

Fig. 14 illustrates the five primary content categories of affected hosted domains for each ISP. It is evident that "Business" and "Information Technology" domains rank among the top five categories across all ISPs, which is similar to Fig. 10, indicating that significant websites are indeed impacted by this type of censorship.

7. Discussion

The results of this study highlight a key structural characteristic of Internet censorship in Iraq. While the MoC establishes regulatory mandates, the absence of a unified technical enforcement mechanism—such as nationwide Deep Packet Inspection (DPI) or a standardized filtering infrastructure—results in heterogeneous implementation across ISPs. This hybrid model provides an important explanation for the observed dichotomy between consistent DNS-based censorship and highly variable TCP/IP blocking.

The relative uniformity of DNS-based censorship across ISPs suggests that policy directives are effectively translated into practice at the resolver level, likely due to the low cost and operational simplicity of DNS interference. However, the variation in DNS response behaviors (e.g., NXDOMAIN, SERVFAIL, empty responses) indicates that

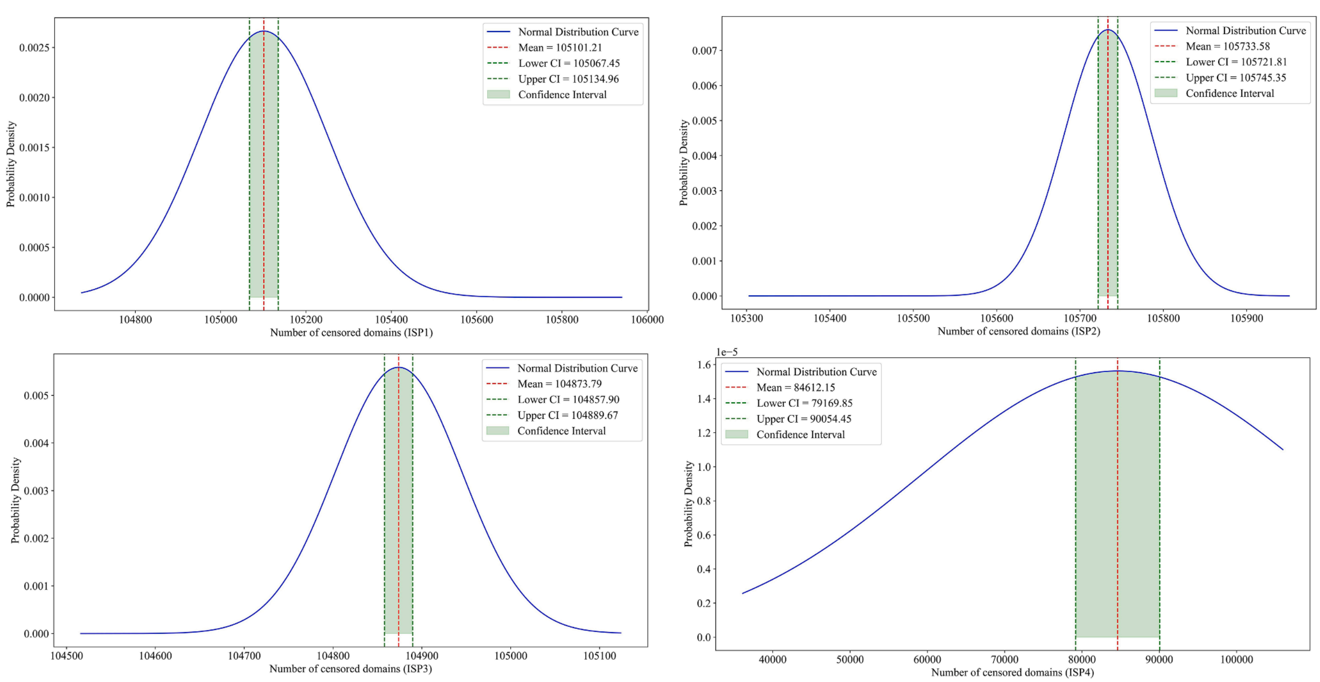

Fig. 11. Number of DNS-based censored domains expressed as a CI (private test list scenario).

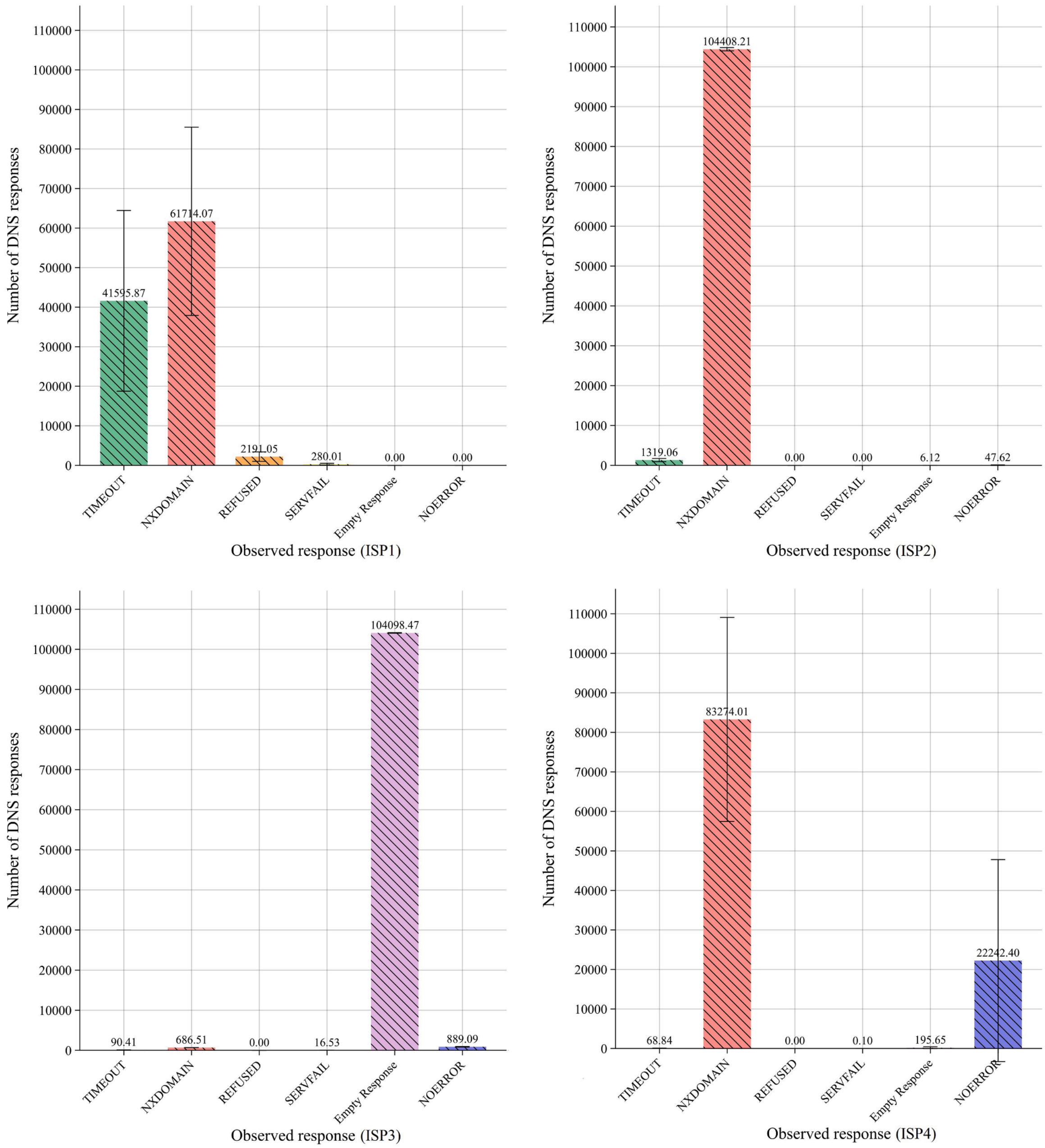

Fig. 12. Observed DNS responses for each ISP.

implementation remains technically decentralized, with each ISP configuring its own resolver infrastructure. This reflects a model in which compliance is achieved through outcome alignment (blocking domains), rather than through standardized technical means.

In contrast, TCP/IP blocking reveals substantial discrepancies among ISPs, indicating that this layer of censorship is largely governed by ISP-specific policies rather than strict regulatory enforcement. The case of ISP2—exhibiting significantly higher levels of IP and domain blocking—illustrates this divergence. Several factors may contribute to this behavior. First, as one of the major providers with a large subscriber base, ISP2 may face greater regulatory scrutiny and therefore adopt over-compliance strategies. Second, larger ISPs typically possess more advanced technical capabilities, enabling them to implement broader or more aggressive filtering. Third, institutional proximity to regulatory authorities may incentivize stricter enforcement. By contrast, ISP1 (a newer provider) appears to implement minimal independent filtering and primarily adheres to baseline regulatory requirements, while ISP3 demonstrates region-specific policy interpretation, particularly in Baghdad. These findings suggest that market position, infrastructure maturity, and regulatory relationships all influence censorship behavior in decentralized environments.

When situated within the broader literature on decentralized censorship, Iraq exhibits both similarities and distinctions. In Russia, for example, earlier decentralized filtering practices have increasingly been replaced by centralized enforcement through systems such as TSPU, leading to more uniform censorship across ISPs. Iraq, by comparison, represents a less mature stage of this evolution, where policy centralization has not yet been matched by technical standardization. In India, another decentralized case, prior studies report significant variation across ISPs but limited reliance on TCP/IP blocking. The Iraqi case differs in that IP-level blocking plays a more prominent role, amplifying the risk of collateral damage. These comparisons position Iraq as a distinct hybrid model in which inconsistent enforcement is not merely incidental but structurally embedded.

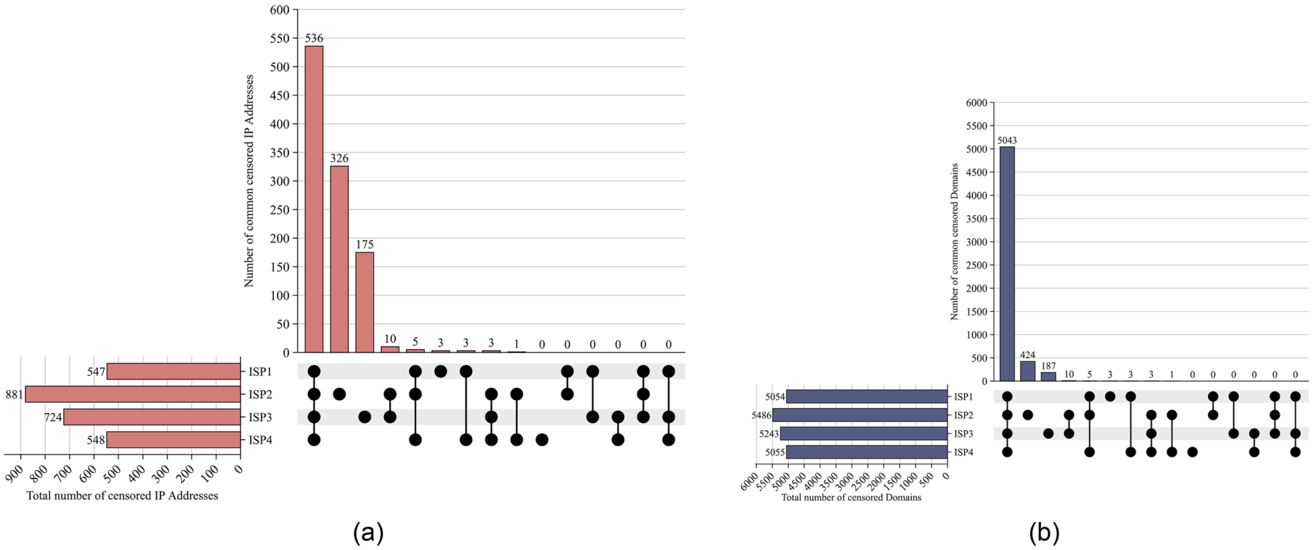

Fig. 13. Overlapping between ISPs (private test list scenario): (a) IP-blocked addresses; (b) Domains affected by IP level blocking.

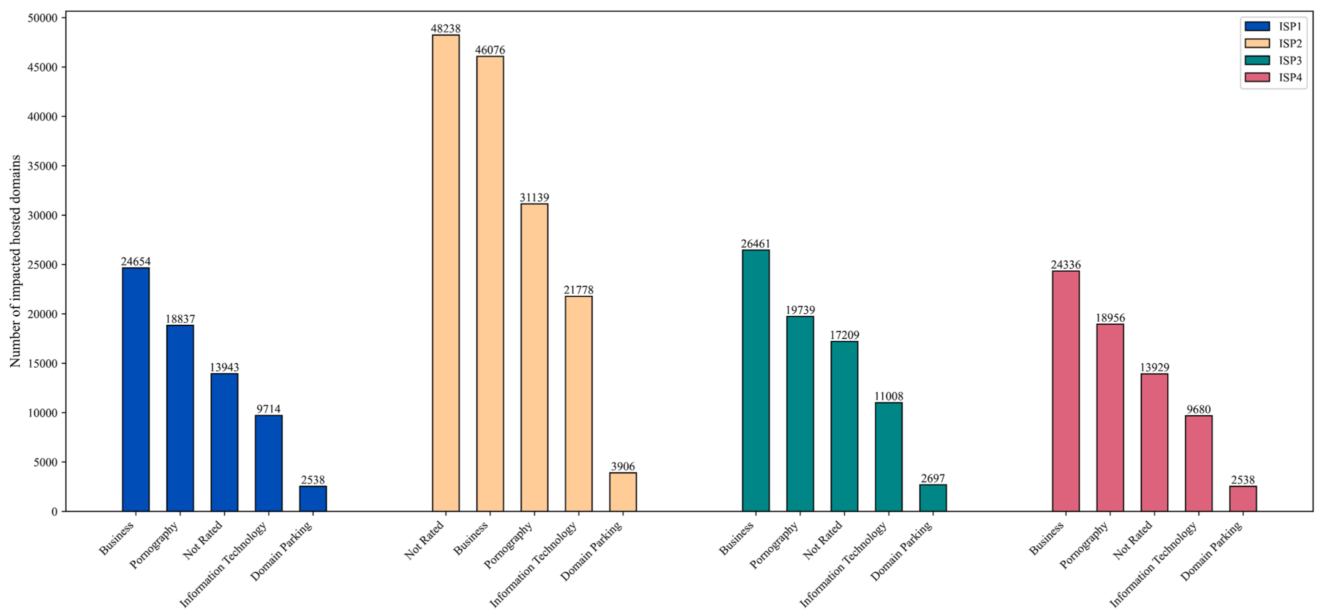

Fig. 14. Top 5 categories of impacted hosted domains for each ISP (private test list scenario).

A particularly important implication of this study is the prevalence of over-blocking associated with TCP/IP filtering. Because multiple domains often share a single IP address, blocking at the IP level can unintentionally restrict access to unrelated services. Our findings show that this collateral impact disproportionately affects domains in the business, information technology, and education sectors, which are critical for economic development and knowledge access. The continued blocking of widely used platforms such as archive.org exemplifies how coarse-grained filtering mechanisms can undermine legitimate use cases. Similar over-blocking phenomena have been documented in highly restrictive environments such as China and Turkmenistan, but remain underexplored in decentralized contexts. This study contributes to filling that gap.

Beyond technical considerations, the findings reveal tangible real-world consequences for Iraqi society and economy. At the macro level, repeated Internet shutdowns have resulted in substantial economic losses, underscoring the direct link between connectivity restrictions and national productivity. At the micro level, over-blocking and inconsistent filtering affect multiple user groups differently. Businesses may experience disruptions to digital services and international platforms, limiting growth and innovation. Educational institutions and students face restricted access to online resources, hindering learning opportunities. Moreover, the blocking of widely used informational platforms has generated skepticism among users, particularly those with technical expertise, raising concerns about transparency, proportionality, and trust in regulatory decisions. These impacts show that censorship operates not only as a technical control mechanism, but also as a policy practice with broader social and economic consequences.

The study also highlights a critical governance gap between policy intent and implementation outcomes. While centralized mandates aim to regulate specific categories of content, the lack of standardized enforcement mechanisms leads to inconsistent application and unintended side effects. Addressing this gap requires moving beyond high-level directives toward concrete regulatory and technical interventions. First, regulators should mandate transparency by requiring ISPs to publish periodic reports detailing blocked domains and the justification for blocking. Second, IP-level blocking should be subject to stricter oversight, with clear justification requirements to minimize arbitrary or excessive filtering. Third, policymakers should prioritize domain-level filtering over IP-level approaches, as it offers greater precision and reduces collateral damage. Fourth, the development of unified technical guidelines or shared filtering frameworks would help ensure more consistent implementation across ISPs. Fifth, establishing protected whitelists for critical sectors—such as education, government services, and business infrastructure—would mitigate the negative impact of over-blocking. Finally, introducing audit and appeal mechanisms would allow affected entities to challenge unjustified blocking, improving accountability.

In summary, this study demonstrates that decentralized enforcement in the absence of technical standardization leads to uneven censorship, increased over-blocking, and measurable societal impact. Iraq's current model underscores the limitations of relying solely on regulatory mandates without coordinated technical implementation. For policymakers, the key challenge is not only defining what should be censored, but also how censorship should be implemented in a manner that is transparent, proportionate, and economically sustainable.

8. Conclusion

This paper examined how content filtering is implemented in Iraq's residential Internet, a setting where regulatory authority and network architecture create a fragmented, decentralized environment. Using repeated active measurements from four residential networks in 2025, we compared outcomes for a popularity-based domain list (Majestic Million) and a government-sourced list from the Ministry of Communications (MoC).

Across both scenarios, DNS interference is the dominant and most uniform technique. For the Majestic list, 386 domains were consistently blocked across all four networks, including highly ranked sites, indicating that filtering is not limited to low-traffic content. IP-level blocking is also present but markedly less consistent across providers: 195 blocked IP addresses were common to all networks, while one major ISP applied many additional IP and domain blocks. Because IP addresses often host large numbers of unrelated services, this approach produces substantial over-blocking; our mapping shows that collateral impact extends beyond the intended categories and disproportionately affects business, information-technology, and education domains.

For telecommunications policy and regulation, these findings highlight a governance gap between centralized directives and operator-level implementation. Where filtering is mandated, policymakers and regulators should prioritize proportionality and auditability: publish or at least standardize blocking rules across operators, favor domain-level controls over IP-level blocks where feasible, and establish review and redress mechanisms to minimize unintended harm to lawful services. Future work can expand coverage to additional regions and enterprise networks, and add application-layer validation (e.g., HTTP/TLS-based tests) to strengthen attribution and further distinguish ISP filtering from remote geo-restrictions.

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Data availability statement

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Ameer Al-Dujaily: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bahaa Al-Musawi:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Ameer Al-Dujaily reports equipment, drugs, or supplies was provided by Iraqi Ministry of Communications (MoC). Bahaa Al-Musawi reports equipment, drugs, or supplies was provided by IPinfo. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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